

Dynamic Fidelity and Model Eligibility in Daily PM₁₀ Forecasting: A Multi-Station Decision Audit

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Abstract

Error-based forecast evaluation can identify improvements over persistence without showing whether forecasts preserve dynamic PM₁₀ behaviour. We audit whether adding explicit dynamic-fidelity requirements changes model eligibility in a rolling-origin benchmark of daily PM₁₀ forecasts from 17 Spanish monitoring stations, five model families, and seven horizons. Rule A requires positive persistence-relative RMSE skill and a significant comparison with persistence; 277 of 595 cells satisfy it. Rule B adds variance retention and P75 exceedance-recall thresholds, leaving 8 eligible cells. Thus, 269 cells change status, corresponding to 97.1% of Rule-A eligible cells. The result shows that error eligibility and dynamic fidelity provide complementary information in this benchmark. Thresholds are operational audit criteria rather than universal standards, and the structured cells are not independent replicates. The conclusion is bounded to the evaluated pollutant, stations, models, horizons, and validation design.

Keywords: PM₁₀ forecasting, model eligibility, dynamic fidelity, forecast evaluation, variance retention, rolling-origin audit

1. Introduction

Environmental forecasting supports decisions that depend on both expected error and the timing or amplitude of changing conditions. A candidate forecast is therefore commonly compared with a meaningful reference forecast, rather than judged only by an isolated error value. Persistence-relative skill provides such a comparison for daily time series, while formal predictive-accuracy tests can assess whether an observed loss difference is statistically distinguishable from the reference [1, 2].

These error-based summaries are necessary but not exhaustive. A point forecast may reduce average squared error by attenuating fluctuations, particularly when the future is uncertain. Lower error can therefore coexist with a forecast trajectory whose variance or episode amplitude is smaller than that of the observations. Forecast verification distinguishes accuracy, association, calibration, and related properties because no single summary answers all of these questions [3, 4].

The distinction matters when evaluation results are converted into eligibility decisions. A cell that improves on persistence under RMSE may remain unsuitable for a decision that requires sensitivity to elevated pollution or preservation of dynamic spread. Variance and event diagnostics are established complements to error measures; they are not introduced here as new metrics. The unresolved practical issue is whether adding explicit fidelity requirements to an error-based eligibility rule materially changes the set of cells that pass.

The EMS audit motivating this study supports a deliberately bounded gap. Existing environmental forecasting studies commonly evaluate predictive error and, less consistently, dynamic-fidelity dimensions, while

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evidence remains limited on whether explicitly combining these dimensions in an eligibility rule materially changes decisions across sites, model families, and forecast horizons. This statement describes the audited corpus; it does not claim that no such study exists elsewhere.

We ask: *Does incorporating dynamic-fidelity criteria alongside conventional error-based statistical evaluation materially alter model eligibility in multi-step daily PM_{10} forecasting?* We answer with a post-evaluation audit of a fixed multi-station benchmark. The contributions are:

1. an empirical audit of error–fidelity discordance;
2. a characterisation of its variation across model families and horizons; and
3. a quantification of how explicit fidelity requirements alter eligibility.

2. Evaluation Context and EMS Gap

Reviews of air-quality forecasting report extensive use of RMSE, MAE, correlation, and related point-error summaries when comparing statistical and machine-learning forecasts [5, 6, 7]. These measures are useful for assessing predictive error, but reporting of forecast variability and event sensitivity is less consistent. The distinction is especially relevant for daily pollution series, in which an average-error improvement may not describe whether elevated episodes retain their observed amplitude.

Rolling-origin validation is an established way to respect temporal order in autoregressive forecasting and reduce leakage risk [8, 9]. It does not, however, determine which diagnostics should define eligibility after the forecasts are generated. A study can be temporally careful and still use an error-only filter. The EMS corpus audit therefore focuses on a narrow decision gap: not whether variance or event metrics exist, but whether their explicit combination with error eligibility is shown to change the set of passing cells across sites, model families, and horizons.

The audited corpus contains studies that report error, dynamic, event, or statistical-comparison dimensions in different combinations. That heterogeneity supports the motivation for a transparent audit, but it is not treated as proof of absence from the wider literature. Paper A contributes the empirical decision audit in the benchmark below; it does not claim that variance retention, exceedance recall, or multi-dimensional verification are novel measurements.

3. Eligibility Audit Framework

The analysis applies two post-evaluation filters to every station–model–horizon cell. They do not produce a final winner for each station and horizon; they quantify how many cells pass a conventional error screen and how many remain after adding dynamic-fidelity requirements.

3.1. Error-based eligibility: Rule A

$$\textbf{Rule A:} \quad \text{Skill} > 0 \quad \text{and} \quad \text{DM}_{BH} \text{ significant.} \quad (1)$$

Skill is persistence-relative RMSE skill. The significance flag comes from a two-sided Harvey–Leybourne–Newbold modified Diebold–Mariano comparison with Benjamini–Hochberg adjustment. The multiplicity family is verified in the implementation as the valid model–horizon comparisons within each dataset; it is not one pooled family across all 595 cells [2, 10].

3.2. Fidelity-aware eligibility: Rule B

$$\textbf{Rule B:} \quad \text{Rule A} \quad \text{and} \quad \alpha \geq 0.50 \quad \text{and} \quad \text{Recall}_{P75} \geq 0.20. \quad (2)$$

The thresholds are operational criteria for this audit. They are not estimated optima, universal standards, or regulatory recommendations. A changed cell satisfies Rule A but fails at least one Rule-B condition.

3.3. Formal discordance

A formal discordant cell satisfies Rule A, has $\text{Recall}_{P75} \geq 0.20$, and has $\alpha < 0.50$. This definition is stricter than the two-dimensional region of positive skill and low alpha shown in Figure 2; the latter does not itself include the DM and recall conditions.

4. Methods: Empirical Benchmark

4.1. Data and study population

The frozen empirical evidence comprises daily PM_{10} series from 17 heterogeneous Spanish monitoring stations identified in the MITECO/EMEP metadata. The station metadata represents urban, traffic, industrial, suburban, and rural settings. The canonical evidence does not establish a single defensible calendar endpoint for the complete recovered station set; consequently, the manuscript reports the verified station coverage and daily resolution without inventing a period label.

The unit of analysis is a station–model–horizon cell. The complete grid contains 17 stations, five empirical model families, and seven horizons, giving 595 rows. Each row summarises matched out-of-sample diagnostics for one model at one station and horizon. Shared stations, time series, and model families mean that these rows are structured cells, not independent replicates.

4.2. Forecast task and validation

The task is direct multi-step daily point forecasting at horizons $h = 1, \dots, 7$. Evaluation follows five expanding rolling-origin folds in chronological order. Information available after an origin is excluded from fitting, feature construction, preprocessing, threshold estimation, and verification inputs for that origin. Missing verification targets are excluded rather than fabricated. The frozen aggregate does not recover exact per-fold row counts or a single complete train/test calendar, so those details are not asserted.

4.3. Leakage controls

The protocol requires train-only treatment of quantities estimated within a split. This includes preprocessing statistics, event thresholds, and model fitting. P75 thresholds are estimated from the corresponding training window rather than from verification outcomes. Persistence is evaluated on the same matched forecast cases used for candidate-model comparisons. The available provenance supports these controls, but does not support additional claims about unrecorded feature or hyperparameter details.

4.4. Model families

The five empirical model labels are direct Histogram Gradient Boosting (HGB), direct Ridge regression, fixed-order SARIMA, seasonal naive, and STL+Ridge. HGB and Ridge are direct horizon-specific forecasters; SARIMA is a recursive statistical family; seasonal naive is a seasonal carry-forward reference; and STL+Ridge is a decomposition/regression hybrid. The verified and unresolved configuration details are consolidated in the model evaluation specification above.

5. Verification Metrics and Statistical Procedures

5.1. Error and baseline-relative skill

For matched forecast–observation pairs, RMSE is the square root of mean squared forecast error. Persistence-relative skill is

$$\text{Skill} = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}}. \quad (3)$$

Positive skill indicates lower RMSE than persistence; it does not establish dynamic fidelity.

Model evaluation specification. Reconstructed from the frozen evidence and repository configuration. “Not recoverable” denotes a detail not established by the canonical provenance; it is not inferred from model names.

Model	Inputs	Forecast strategy	Fold fitting	Verified configuration	Known limitation
HGB direct	Direct lag-based predictors; complete feature list not recoverable	One direct forecast per horizon	Fit within rolling-origin training data	Family label and 7-horizon evaluation verified	Exact hyper-parameters not recoverable from frozen provenance
Ridge direct	Direct lag-based predictors; complete feature list not recoverable	One direct forecast per horizon	Fit within rolling-origin training data	Family label and 7-horizon evaluation verified	Exact regularisation and feature list not recoverable
SARIMA	Daily PM ₁₀ series	Recursive statistical forecast	Fit within rolling-origin training data	Fixed-order SARIMA family verified	Historical configuration records conflict; exact order omitted
Seasonal naive	Historical seasonal observations	Seasonal carry-forward baseline	No parameter fitting beyond training history	Family label and 7-horizon evaluation verified	Seasonal period details not recoverable from canonical table
STL+Ridge	Decomposition and regression inputs	Hybrid decomposition/regression forecast	Fit within rolling-origin training data	Family label and 7-horizon evaluation verified	Exact STL and Ridge settings not recoverable

5.2. Variance retention and event recall

The canonical dynamic diagnostic is the predicted-to-observed variance ratio

$$\alpha = \frac{\text{Var}(\hat{y})}{\text{Var}(y)}, \quad (4)$$

computed with population variances (ddof = 0) over matched cell pairs. It is a variance ratio, not a standard-deviation ratio. P75 recall is the proportion of observed P75 exceedances correctly identified by the forecast, with the threshold estimated from the corresponding training window.

5.3. Diebold–Mariano testing and multiplicity

The loss differential compares squared errors from the candidate forecast and persistence. The two-sided Diebold–Mariano statistic uses the Harvey–Leybourne–Newbold small-sample correction and horizon-dependent autocovariance truncation. Benjamini–Hochberg adjustment is applied within each dataset to the valid model–horizon p-values before the significance flag is defined at 0.05. This family definition is important because the 595 cells share data and are not treated as independent tests in a single inferential population.

5.4. Bootstrap uncertainty

The canonical variance diagnostic includes descriptive percentile-bootstrap intervals computed with 1,000 resamples and seed 42. These intervals describe uncertainty around cell-level alpha estimates; they are not used to optimise Rule-B thresholds and are not interpreted as population-level inference over the structured grid.

6. Evidence Governance and Reproducibility

The empirical analysis is governed by a frozen 595-row diagnostic table whose SHA-256 is recorded in the provenance manifest. The table contains the complete five-model, 17-station, seven-horizon grid and is the sole source for the numerical results, tables, and primary figures in this manuscript.

Reproducibility is separated into two layers. The scientific layer defines the forecast task, validation controls, metrics, eligibility rules, and limitations in the sections above. The computational layer preserves source tables, provenance records, deterministic table and figure generators, and a claim-level traceability matrix in the repository package. The final submission archive contains only LaTeX inputs and figures needed for compilation; internal audit material remains outside that archive.

The sensitivity analysis is a deterministic transformation of the frozen rows. It changes neither forecasts nor thresholds used for the primary result. Its purpose is to show how the eligibility consequence behaves in a prespecified neighbourhood of the operational thresholds.

7. Results

The results follow the eligibility audit: error-based eligibility, dynamic fidelity, formal discordance, the primary eligibility consequence, and deterministic threshold sensitivity. Table 1 reports model-level medians and counts; Table 2 reports the primary rule comparison.

7.1. Error-based eligibility

Rule A identifies 277 eligible cells: 111 HGB, 117 Ridge, 44 SARIMA, 5 seasonal naive, and 0 STL+Ridge. These are eligibility counts within the benchmark, not population rankings and not evidence that one model is the final winner for every station and horizon.

Table 1: Per-model diagnostic summary across 595 station–model–horizon cells (17 stations $\times$ 5 models $\times$ 7 horizons). Skill is persistence-relative RMSE skill; α is variance retention; DM sig. counts BH-adjusted Diebold–Mariano cells significant at $p < 0.05$; Rule A: skill > 0 and DM significant; Rule B: Rule A and $\alpha \geq 0.5$ and recall_{P75} ≥ 0.20 ; Discordant: Rule A, recall_{P75} ≥ 0.20 , and $\alpha < 0.5$. Medians and means computed over all cells per model. Cells are not independent (shared stations and series); the central tendency columns are medians and the remaining columns are counts.

Model	n	Median Skill	Median α	Collapse	DM sig.	Rule A	Rule B	Discordant
HGB direct	119	0.205	0.151	118/119	111/119	111/119	1/119	54/119
Ridge direct	119	0.219	0.087	118/119	117/119	117/119	1/119	43/119
SARIMA	119	0.208	0.095	110/119	44/119	44/119	1/119	4/119
Seasonal naive	119	-0.026	1.000	0/119	39/119	5/119	5/119	0/119
STL+Ridge	119	-1.107	1.399	0/119	119/119	0/119	0/119	0/119

7.2. Dynamic-fidelity behaviour

Variance collapse ($\alpha < 0.50$) occurs in 118/119 HGB cells, 118/119 Ridge cells, and 110/119 SARIMA cells. Seasonal naive has 0/119 collapsed cells, while STL+Ridge has 0/119 collapsed cells and a median alpha above one. Figure 1 shows how the median skill and alpha profiles vary with horizon. These are descriptive patterns within the evaluated benchmark and do not establish an architectural or causal mechanism.

7.3. Error-fidelity discordance

Figure 2 shows the joint diagnostic. Across all 595 cells, Spearman correlation between alpha and skill is $\rho = -0.863$. This pooled association is descriptive and partly reflects differences among model-family profiles; it should not be interpreted as a homogeneous relationship within every family or horizon.

There are 101 formal discordant cells: 54 HGB, 43 Ridge, and 4 SARIMA. They satisfy Rule A, have P75 recall of at least 0.20, and have $\alpha < 0.50$. Thus, within this benchmark, positive error eligibility and adequate P75 recall can coexist with low retained variance.

7.4. Effect on model eligibility

Adding the fidelity requirements changes eligibility from 277 cells under Rule A to 8 under Rule B: HGB 111 $\rightarrow$ 1, Ridge 117 $\rightarrow$ 1, SARIMA 44 $\rightarrow$ 1, seasonal naive 5 $\rightarrow$ 5, and STL+Ridge 0 $\rightarrow$ 0. The 269 changed cells are $269/277 = 97.1\%$ of Rule-A eligible cells.

Horizon-wise error and fidelity profiles

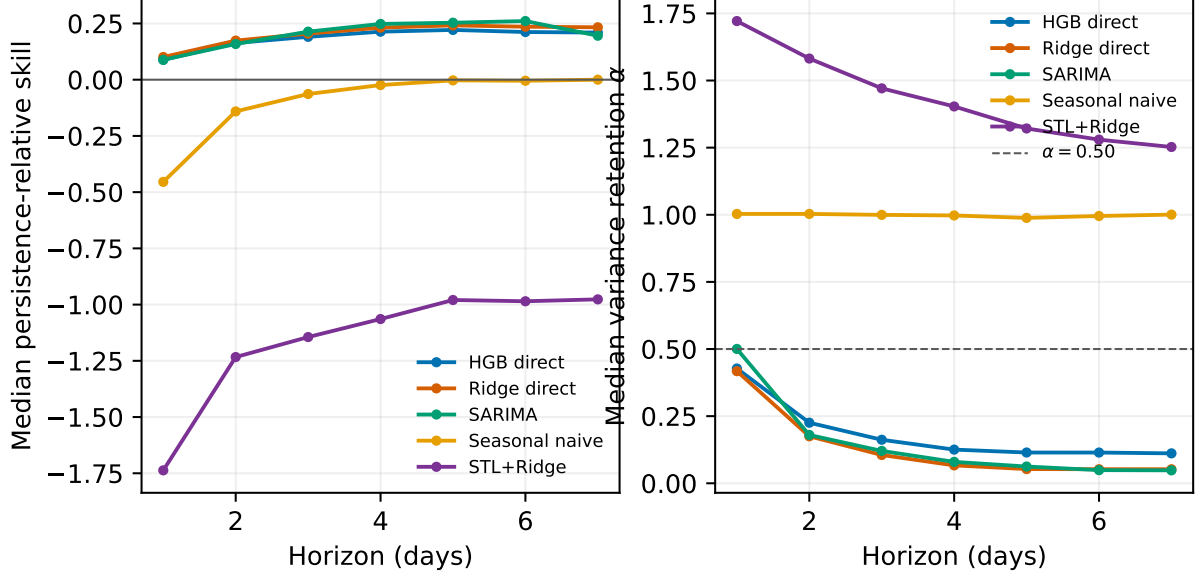


Figure 1: Median persistence-relative skill and median variance retention by horizon and model family. Each point summarises the 17 station cells at that horizon.

7.5. Rule-B threshold sensitivity

The primary result uses $\alpha \geq 0.50$ and $P75 \text{ recall} \geq 0.20$. A deterministic sensitivity audit over the neighbouring grid $\alpha \in \{0.40, 0.50, 0.60\}$ and $\text{recall} \in \{0.10, 0.20, 0.30\}$ retains the qualitative conclusion: 91.3–98.2% of Rule-A cells change status across the tested neighbourhood, with the exact values reported in Table 3. The primary point remains 8 retained cells and 97.1% changed. This is robustness of the decision consequence within the tested operational neighbourhood, not evidence that either threshold is optimal.

8. Discussion

8.1. The main decision consequence

The central result is not the introduction of α as a metric. It is the change in the set of eligible forecast cells when dynamic fidelity is added to an error-based screen. In this benchmark, 277 cells pass Rule A and only 8 pass Rule B, so 269 Rule-A cells fail at least one stated fidelity condition. The result makes the practical consequence of multi-dimensional verification explicit: the answer to “does this cell improve on persistence?” is not equivalent to the answer to “does this cell also meet the selected dynamic-fidelity requirements?”

8.2. Error and fidelity are complementary

RMSE remains an appropriate and necessary summary of point-forecast error. The audit does not reject it or imply that lower RMSE is uninformative. It shows instead that RMSE skill is a different property from the variance and episode behaviour represented by α and P75 recall. A model can reduce the magnitude of many errors while attenuating the spread of its trajectory. Conversely, a forecast can retain spread while mistiming or mis-sizing events. Reporting these dimensions together clarifies which requirement causes a cell to pass or fail.

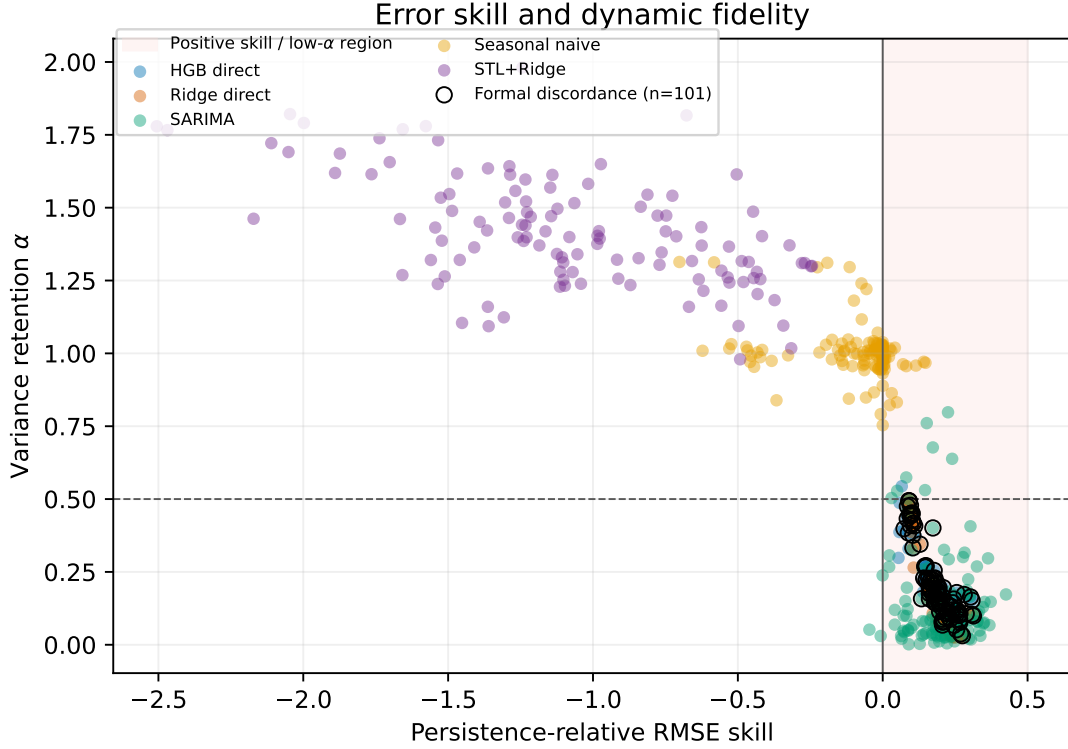


Figure 2: Persistence-relative RMSE skill versus canonical variance retention α across 595 cells. The shaded region represents only positive skill with $\alpha < 0.50$; it is not the formal discordance definition. Black open contours identify the 101 cells that additionally satisfy Rule A and P75 recall ≥ 0.20 .

8.3. Model-family profiles

The model-family profiles illustrate several trade-offs without establishing architectural causality. HGB, Ridge, and SARIMA contribute many Rule-A cells but also show frequent low-alpha cells. Seasonal naive retains variance in this diagnostic but contributes few error-eligible cells. STL+Ridge avoids low alpha at the point-estimate level but has poor persistence-relative skill and therefore contributes no Rule-A cells. The appropriate interpretation is not that one family is universally preferable, but that different families occupy different positions in the joint eligibility space.

8.4. Relation to forecast verification

The EMS audit does not suggest that variance diagnostics, event metrics, or multi-dimensional verification are new inventions. Existing forecast-verification work emphasises that accuracy and other properties should be interpreted jointly, and air-quality reviews document the prevalence of error-based reporting [3, 4, 5, 6, 7]. The contribution of this paper is narrower: it quantifies the eligibility consequence of adding explicit fidelity conditions to an error-based rule over a structured multi-station, multi-model, multi-horizon benchmark.

8.5. Threshold sensitivity and operational interpretation

The deterministic sensitivity audit shows that the large change in eligibility persists in the tested neighbourhood around alpha 0.50 and P75 recall 0.20. This supports robustness of the qualitative decision consequence, not optimality of the selected thresholds. In an operational application, preserving dynamic spread or detecting elevated episodes may matter when forecasts inform alert thresholds, episode planning, or amplitude-sensitive decisions. The benchmark does not measure the value of any specific operational intervention, so these implications remain conditional.

Table 2: Effect of adding dynamic-fidelity requirements (Rule B) to error-based eligibility (Rule A) across the 595-cell empirical benchmark. Rule A: skill > 0 and BH-adjusted DM significant; Rule B: Rule A and $\alpha \geq 0.5$ and $\text{recall}_{P75} \geq 0.20$. Decision changes indicate the number of cells eligible under Rule A but not Rule B. Cells are not independent observations.

Model	Rule A	Rule B	Changes
HGB direct	111	1	110
Ridge direct	117	1	116
SARIMA	44	1	43
Seasonal naive	5	5	0
STL+Ridge	0	0	0
Total	277	8	269 (97.1%)

Table 3: Deterministic sensitivity of Rule B around the primary thresholds. Each cell reports retained Rule-B eligible cells and, in parentheses, the percentage of Rule-A cells that change status. The 595 canonical diagnostic rows are reused; no forecasts are recomputed.

α threshold \ recall threshold	0.10	0.20	0.30
0.40	24 (91.3%)	24 (91.3%)	23 (91.7%)
0.50	8 (97.1%)	8 (97.1%)	7 (97.5%)
0.60	6 (97.8%)	6 (97.8%)	5 (98.2%)

8.6. Limitations and boundaries

The evidence is bounded to daily PM_{10} from 17 Spanish monitoring stations, five model families, seven horizons, and five expanding rolling-origin folds. The 595 cells share stations and underlying series and are not independent replicates. Rule-B thresholds are study-specific. The analysis uses deterministic point forecasts, establishes no causal mechanism for low α , and has limited transferability to other pollutants, countries, temporal resolutions, horizons, or model families. The aggregate evidence is recovered and frozen, and the row-level provenance notes a specific limitation in independently reconstructing SARIMA skill from the combined row-level release. Historical SARIMA configuration records conflict; the manuscript therefore does not assert an exact order beyond the fixed-order family description supported by the evaluation package.

9. Conclusions

In the evaluated multi-station daily PM_{10} benchmark, 277 of 595 cells satisfied conventional error-based eligibility, whereas 8 satisfied the same rule plus the stated dynamic-fidelity requirements. The 269 changed statuses represent 97.1% of Rule-A eligible cells, and 101 cells met the stricter formal discordance definition.

The methodological implication is that persistence-relative error skill and dynamic fidelity should be reported as complementary eligibility dimensions. The conclusion is descriptive and conditional on the pollutant, stations, model families, horizons, validation design, and study-specific thresholds; it is not a universal claim about forecasting systems.

CRediT author statement

Federico García Crespi: Conceptualisation, Data curation, Formal analysis, Investigation, Methodology, Software, Visualisation, Writing – original draft, Writing – review & editing. **Julio Alberto Ramos Martínez:** Conceptualisation, Supervision, Writing – review & editing.

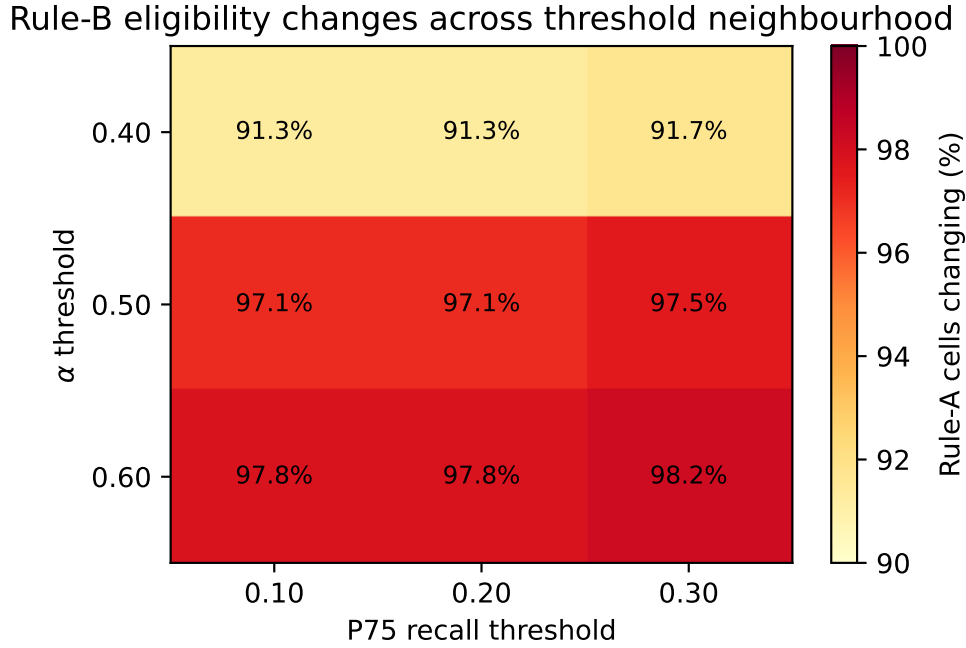


Figure 3: Deterministic sensitivity of the proportion of Rule-A eligible cells that change status when the Rule-B alpha and P75-recall thresholds vary around their primary values.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Software Availability

The analysis repository is maintained at <https://github.com/fedeg-umh-es/varret-pm10-paper>. The canonical aggregate diagnostic table, provenance manifest, reproducible table and figure generators, and documented reconstruction limitations are included in the repository package. The empirical evidence identifies 17 Spanish MITECO/EMEP monitoring stations and contains 595 station-model-horizon rows. The repository must be synchronised with the final submission commit before an EMS submission; no external DOI is asserted here.

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Supplementary Material

S1. Monitoring stations

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Table 4: Monitoring stations represented in the recovered station metadata. The table is descriptive and does not introduce additional empirical rows beyond the canonical 595-cell benchmark.

Station code	Name	Area type	Altitude (m a.s.l.)	Coverage (%)
ES0446A	Elx-Agroalimentari	Suburban/Industrial	44	80.4
ES1619A	Valencia-Vivers	Urban/Residential	11	91.7
ES0012R	Zarra-EMEP	Rural Remote EMEP	855	96.0
ES1870A	Barcelona El Port Vell	Suburban/Industrial	1	98.5
ES1418A	Alagon	Suburban/Traffic	235	97.5
ES1421A	Teruel	Urban/Background	915	96.7
ES0001R	San Pablo de los Montes	Rural	923	96.1
		Remote/Background		
ES1438A	Barcelona L'Eixample	Urban/Traffic	26	95.3
ES0567A	Barcelona Zona	Urban/Background	24	95.2
	Universitaria			
ES1417A	Huesca	Urban/Traffic	488	95.0
ES0691A	Barcelona El Poblenou	Urban/Background	3	94.9
ES0559A	Barcelona Pl. Universitat	Urban/Traffic	24	94.9
ES1856A	Barcelona Vall d'Hebron	Urban/Background	136	94.5
ES1966A	Alcaniz Capuchinos	Urban/Industrial	340	94.3
ES2011A	Sant Vicenc dels Horts	Suburban/Industrial	65	94.1
	Alaba			
ES2091A	Alcanar Montecarlo	Rural	6	93.7
		Remote/Industrial		
ES2017A	Alcanar Llar de Jubilats	Rural	7	93.6
		Near-city/Industrial		

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